Project A - System Architecture

1. Overview

Project A is a three-tier web application. The presentation tier is a React single-page application served through a content delivery network. The application tier is a FastAPI backend exposing a REST API. The data tier is a PostgreSQL database with a Redis cache. All components run as containers on a managed Kubernetes cluster in two availability zones.

2. Deployment diagram

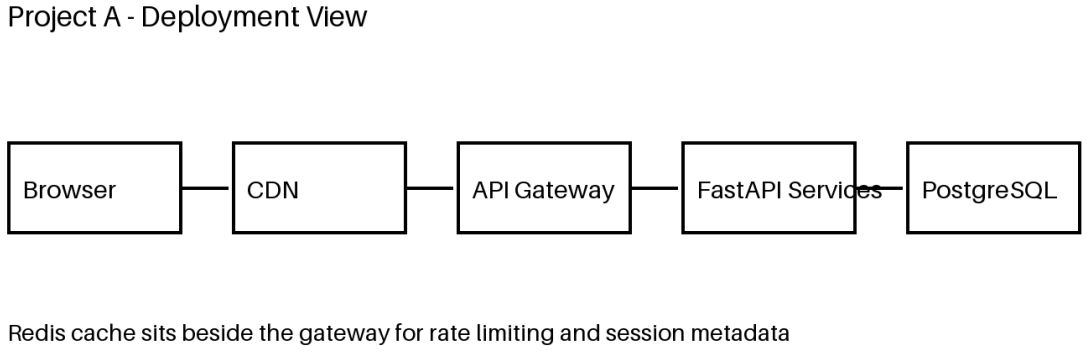

Figure 1: Deployment view showing the request path from browser to database.

3. Component inventory

Component	Technology	Replicas	Purpose
Frontend	React + TypeScript	CDN	Customer web interface
API gateway	Envoy	3	TLS, rate limiting, JWT validation
Account service	FastAPI	4	Balances and transactions
Statement worker	Python + queue	2	Asynchronous PDF statements
Database	PostgreSQL 15	1 primary + 2 read	System of record
Cache	Redis	2	Rate limits, session metadata

4. Scalability and recovery

Load tests showed that four backend replicas and a database with two read replicas sustain 5000 concurrent users at a 95th percentile latency of 1.4 seconds. Recovery time objective is 4 hours and recovery point objective is 15 minutes. Nightly backups use point-in-time recovery.

5. Observability

Structured logs are shipped to a central platform with customer identifiers masked. Alerts page the on-call engineer when the error rate exceeds 2 percent for five minutes.